Parent Messaging (Room Chat) - Parents' view

0) Scope & Assumptions

- **Users:** Parents/Guardians.
- **Purpose:** Enable parents to communicate with teachers/staff regarding their child’s activities, updates, and queries.
- **Tenancy:** Messaging is scoped to the selected organization and school context. But Parent can belong to one or more schools; each parent may have multiple children enrolled.
- **Session:** Browser-based SPA and Mobile App (iOS/Android); JWT or session cookie.
- **Timezone:** Use school’s timezone for timestamps.

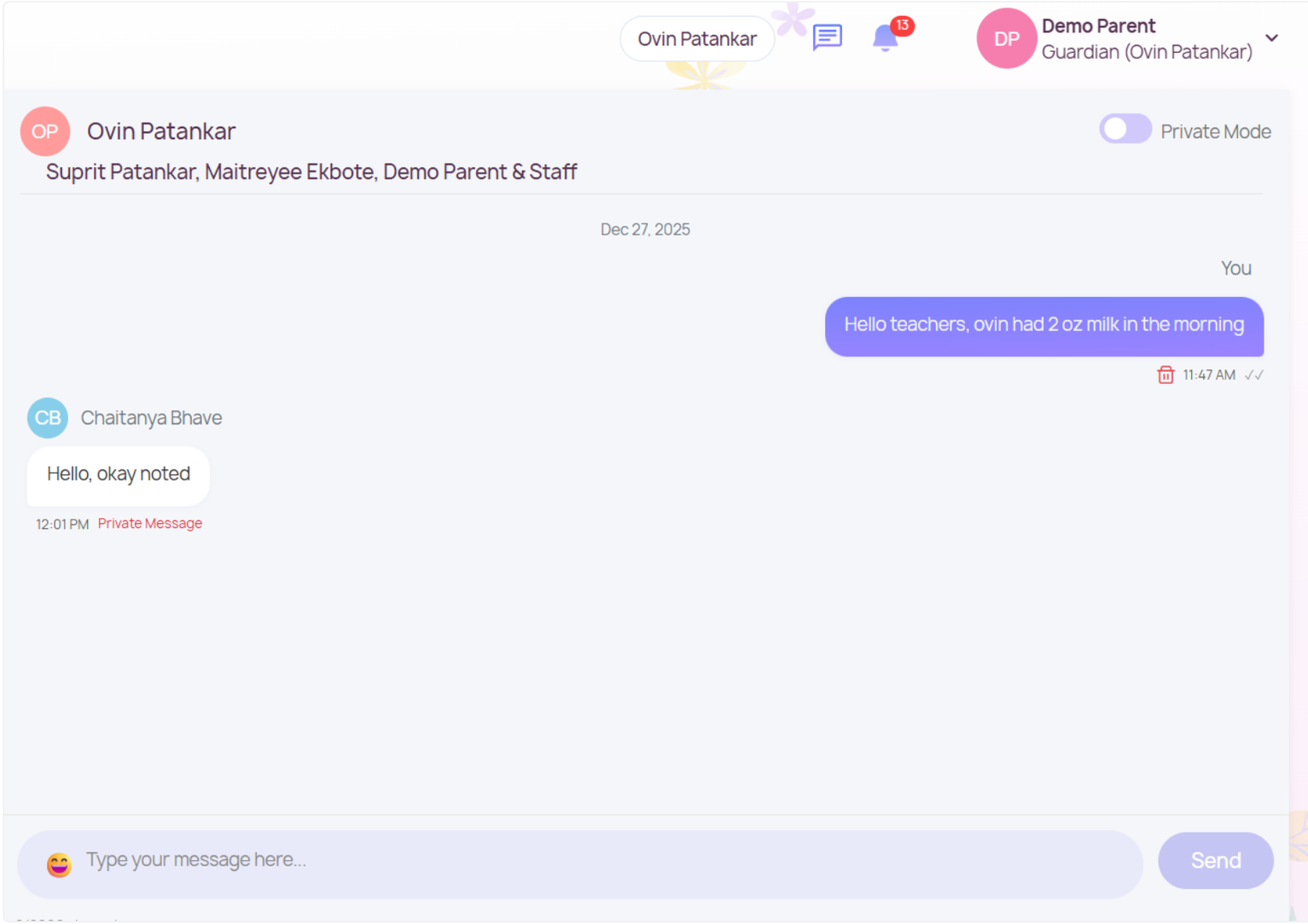
User Journey

Persona: Parent
Actions:

- Login
- Click **Messaging** from menu → **Room Chats (**Directly opens conversation thread for their child (no sidebar).
- If multiple children: Use **Child Dropdown** at top to switch between children.
- Compose and send messages from **web or mobile**.
- View responses from staff/teachers.
- Toggle **Private Mode on** for confidential messages.

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1) UI & Flow



- **Header:**
  - Child's name prominently displayed.
  - Dropdown for child selection (if multiple children).
  - Participant list (teachers, staff).
  - Notification badge for unread messages.
  - Private Mode toggle.
- **Conversation Thread:**
  - Messages displayed in chronological order.
  - Parent messages styled distinctly (e.g., purple bubble, "You" label).
  - Staff messages show sender name and role.
  - Private messages show "Private" indicator.
  - Message status: ✓ Sent, ✓✓ Delivered Blue tick is Read.
  - **No chat sidebar showing chat titles with search & filter option for parent view**
- **Composer:**
  - Placeholder: "Type your message here...".

- Emoji picker.
- Rich text formatting (bold, italic, underline).
- URL-compatible text.
- Attachement- Same as Admin/ staff
- Send button (disabled when empty).

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2) Field Specifications

Child Dropdown

Field Name	Data Type	Default Value	Validations	Notes
Child Selection	Enum	First child	Must be active child	Dropdown visible only if >1 child

Message Composer

Field Name	Data Type	Default Value	Validations	Notes
Message Text	String	Empty	Required; Max 2000 chars	Rich text + URL support
Emoji Picker	Boolean	False	N/A	Opens emoji panel
Private Mode	Boolean	OFF	N/A	Toggle for confidential messages, used for seperated parents/ split family
Send Button	Boolean	Disabled	Enabled only if text entered	Primary CTA
Attachement	file	None	optional, JPEG, PNG, HEIC, PDF & Docs files allowed each file<10mb,	Photo visible directly in chat. , PDF or doc is like file (sameas in whatsapp) multiple

			Multiple allowed	photo or file view same as whatsapp

Acceptance Criteria

1. Conversation View

- **AC1:** Parent sees child’s name and participant list at the top of the conversation.
- **AC2:** Messages are displayed in chronological order with timestamps.
- **AC3:** Parent’s /user's own messages appear in distinct style (e.g., purple bubble) and show “You” label.
- **AC4:** Staff messages display sender name and role clearly.
- **AC5:** Private messages show a “Private” indicator.
- AC6 : Always redirect to recent message in chat

2. Message Composition

- **AC6:** Input box placeholder reads “Type your message here...”.
- **AC7:** Send button is disabled when input is empty.
- **AC8:** Send button becomes enabled when text or emoji is entered.
- **AC9:** Emoji picker opens when emoji icon is clicked.
- **AC10:** Rich text formatting (bold, italic, underline) and clickable URLs are supported.
- **AC11:** Message length cannot exceed 2000 characters; show character counter if needed.

3. Sending & Delivery

- **AC12:** On clicking Send, message appears instantly in thread. Send button should be disabled until action completes
- **AC13:** Sent message shows timestamp and delivery status (✓ sent, ✓✓ read).
- **AC14:** Delivery status updates within 5 seconds after recipient reads message.

4. Private Mode

- **AC15:** Toggle for Private Mode is visible and functional.
- **AC16:** When Private Mode is ON, message is marked as private and only visible to staff (not other parents).
- **AC17:** When Private Mode is OFF, message is visible to all participants.

5. Message Actions

- **AC18:** Parent can delete their own messages within time specified by school in communication settings; confirmation prompt appears before deletion.
- **AC19:** Deleted message is replaced with “Message deleted” placeholder.

- **AC21:** Copy message text option is available.

6. Notifications

- **AC22:** Notification badge shows unread message count accurately.
- **AC23:** Badge updates in real-time when messages are read.

7. Performance & Reliability

- **AC24:** Conversation loads within 500 ms (p95).
- **AC25:** Message send completes within 800 ms (p95).

Additional Acceptance Criteria

- **AC1:** Clicking Messaging menu opens child's conversation thread directly (no sidebar).
- **AC2:** If multiple children, dropdown appears at top; switching updates thread.
- **AC3:** Messages display in chronological order with timestamps. Scroll or redirect to latest message by default automatically
- **AC4:** Parent messages show "You" label and distinct style.
- **AC5:** Staff messages show sender name and role.
- **AC6:** Private Mode toggle marks message as private and restricts visibility to staff only.
- **AC7:** Composer supports rich text and clickable URLs.
- **AC8:** Send button disabled when input is empty; enabled when text or emoji entered.
- **AC9:** Message length capped at 2000 characters.
- **AC10:** Delivery status updates within 5 seconds after recipient reads message.
- **AC11:** Deleted messages replaced with "Message deleted" placeholder.
- **AC12:** Notification badge updates in real-time for unread messages.
- **AC13:** Mobile and web experience consistent; responsive design.

Gherkin Scenarios

Feature: Parent Messaging for Child

Scenario 1: Send a message

```
Given I am viewing my child's conversation thread
When I type "Hello teachers, Ovin had milk" in the input box
And I click Send
Then the message appears in the thread with timestamp
And the delivery status shows one check mark
```

Scenario 2: Send message in Private Mode

```
Given I am viewing my child's conversation thread
And Private Mode is toggled ON
When I type "This is private" and click Send
Then the message appears with a Private indicator
And is only visible to staff participants
```

Scenario 3: Delete a message

```
Given I have sent a message
When I click the Delete icon on that message
And confirm deletion
Then the message is replaced with "Message deleted"
```

Scenario 4: Emoji and Rich Text Support

```
Given I am composing a message
When I click the emoji icon and select an emoji
Then the emoji appears in the input box
And when I apply bold formatting to text
Then the message retains formatting after sending
```

Scenario 5: Message Status Updates

```
Given I have sent a message
When the recipient reads the message
Then the status changes from one check mark to two check marks within 5 seconds
```

Scenario 6: Notifications

```
Given I have unread messages in other threads
When I view the messaging dashboard
Then the notification badge shows the correct unread count
```

Scenario: Open Messaging for Single Child

```
Given I am logged in as a parent with one child
When I click Messaging from the menu
Then the conversation thread for that child opens directly
And no child dropdown is shown
```

Scenario: Switch Child in Dropdown

```
Given I am logged in as a parent with multiple children
When I click Messaging from the menu
Then the conversation thread for the first child opens
And a child dropdown is visible
When I select another child from the dropdown
Then the conversation updates to that child's thread
```

Scenario: Send Message

```
Given I am viewing my child's conversation thread
When I type "Hello teachers" in the composer
And click Send
Then the message appears in the thread with timestamp
And shows one check mark for sent status
```

Scenario: Private Mode

```
Given I am viewing my child's conversation thread
And Private Mode is toggled ON
```

```
When I send a message
Then the message appears with a Private indicator
And is only visible to staff participants & not to other parent / guardian
```

Scenario: Delete Message

```
Given I have sent a message
When I click Delete and confirm
Then the message is replaced with ”Message deleted”
```

Scenario: Rich Text & URL

```
Given I am composing a message
When I apply bold formatting and add a URL
Then the message retains formatting and URL is clickable after sending
```

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5) Security Requirements

- Role-based access: Only parents can view their child’s thread.
- Validate child-parent relationship server-side.
- Audit logs: userId, childId, messageId, timestamp, action (send/delete).

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6) Accessibility & Localization

- Keyboard navigation for composer and dropdown.
- ARIA labels for message status and private toggle.
- WCAG 2.1 AA compliance.
- All strings i18n-ready.

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7) Non-Functional Requirements

- **Performance:** Thread load < 500 ms; message send < 800 ms.
- **Scalability:** Support parents with up to 10 children without UI lag.
- **Reliability:** 99.9% uptime.
